

ESQUIMALT HARBOUR, BC, Canada

WMO: 717980

Lat: 48.432N

Lon: 123.439W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.2	-0.5	-9.7	1.6	0.5	-6.4	2.2	1.5	12.3	8.4	10.7	7.1	4.4	20	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	22.1	15.7	20.5	15.1	19.1	14.7	16.6	20.4	15.9	19.2	15.3	18.2	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.7	17.8	14.4	10.3	17.1	13.9	9.9	16.5	46.6	20.6	44.5	19.4	42.7	18.3	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.4	DB	-4.0	26.7	2.0	1.8	-5.5	28.0	-6.7	29.0	-7.8	30.0	-9.3	31.3	(4)
(5)			WB	-5.2	18.5	2.4	0.9	-6.9	19.2	-8.3	19.7	-9.7	20.3	-11.5	21.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	5.5	5.8	7.1	9.2	11.9	14.1	15.9	15.8	13.7	10.3	7.3	5.3	(6)
(7)		DBStd	4.47	2.74	2.40	2.32	2.21	2.36	2.06	1.80	1.73	1.84	1.95	2.56	2.65	(7)
(8)		HDD10.0	651	140	119	93	40	8	0	0	0	0	20	84	147	(8)
(9)		HDD18.3	2983	398	352	348	275	199	129	78	81	139	249	330	405	(9)
(10)		CDD10.0	717	1	0	3	15	68	122	183	180	112	29	4	1	(10)
(11)		CDD18.3	7	0	0	0	0	0	2	2	2	1	0	0	0	(11)
(12)		CDH23.3	22	0	0	0	0	2	7	6	6	1	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.0	3.2	3.2	3.2	3.2	3.5	3.1	2.7	2.3	2.5	2.9	3.1	(14)	
(15)	Precipitation	PrecAvg	646	107	73	48	34	26	19	14	21	33	61	98	112	(15)
(16)		PrecMax	873	217	171	109	72	52	68	38	67	72	207	191	264	(16)
(17)		PrecMin	389	16	11	12	17	4	2	0	0	4	4	16	10	(17)
(18)		PrecStd	125	53	39	27	14	12	15	11	17	18	46	44	54	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	12.2	14.9	18.5	22.1	23.9	24.0	23.9	22.1	17.4	13.9	11.7	(19)
(20)			MCWB	9.9	8.6	10.0	11.4	14.0	15.8	16.5	16.8	15.5	12.1	11.9	10.3	(20)
(21)		2%	DB	10.6	10.7	12.5	15.2	18.5	20.4	21.8	21.5	19.4	15.1	12.4	10.3	(21)
(22)			MCWB	9.1	8.3	9.2	10.5	12.7	14.6	15.9	15.9	14.7	12.0	10.8	8.9	(22)
(23)		5%	DB	9.7	9.9	11.2	13.5	16.5	18.3	20.2	19.8	17.7	14.0	11.4	9.4	(23)
(24)			MCWB	8.5	7.8	8.7	9.9	12.2	13.8	15.3	15.5	14.1	11.7	10.1	8.1	(24)
(25)		10%	DB	8.9	9.0	10.2	12.2	15.0	16.9	18.8	18.5	16.4	13.0	10.6	8.7	(25)
(26)			MCWB	7.9	7.3	8.2	9.3	11.6	13.1	14.8	14.9	13.6	11.3	9.5	7.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.2	11.1	12.5	15.0	16.8	17.9	17.7	16.7	14.3	12.5	10.8	(27)
(28)			MCDWB	11.3	11.1	12.7	15.9	19.5	22.1	22.0	21.7	20.1	15.3	13.3	11.5	(28)
(29)		2%	WB	9.6	9.2	10.1	11.4	13.6	15.2	16.6	16.6	15.6	13.0	11.3	9.4	(29)
(30)			MCDWB	10.2	10.1	11.5	14.1	17.2	19.0	20.4	20.2	18.3	14.0	11.9	10.1	(30)
(31)		5%	WB	8.8	8.4	9.3	10.6	12.7	14.3	15.8	15.9	14.7	12.2	10.5	8.5	(31)
(32)			MCDWB	9.5	9.4	10.8	12.7	15.6	17.5	19.1	19.0	16.8	13.2	11.2	9.2	(32)
(33)		10%	WB	8.1	7.7	8.6	9.9	12.0	13.6	15.2	15.2	14.0	11.5	9.8	7.8	(33)
(34)			MCDWB	8.9	8.7	10.0	11.8	14.5	16.3	18.1	17.9	15.8	12.6	10.4	8.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.0	4.7	5.2	5.6	6.0	6.0	6.8	6.5	6.0	5.0	4.3	3.9	(35)
(36)			MCDBR	4.7	5.7	6.3	7.9	8.6	8.7	9.0	8.5	8.1	6.1	4.7	4.4	(36)
(37)			MCWBR	3.5	3.8	3.8	4.2	4.0	3.9	4.1	4.0	4.3	3.8	3.4	3.4	(37)
(38)		5% WB	MCDBR	4.3	4.5	5.8	6.8	7.6	8.0	8.4	8.0	7.2	5.2	4.1	4.1	(38)
(39)			MCWBR	3.5	3.5	3.8	4.0	3.9	4.0	4.3	4.1	4.3	3.7	3.4	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.319	0.334	0.362	0.382	0.360	0.351	0.356	0.342	0.335	0.328	0.316	(40)	
(41)		taud	2.516	2.543	2.514	2.425	2.388	2.468	2.530	2.515	2.553	2.557	2.496	2.484	(41)	
(42)		Ebn at Noon	755	847	887	889	880	901	904	886	868	816	733	706	(42)	
(43)		Edh at Noon	62	74	88	106	115	106	99	96	84	72	62	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.93	1.81	2.83	4.29	5.48	5.99	6.55	5.52	3.94	2.17	1.11	0.75	(44)	
(45)		RadStd	0.12	0.24	0.33	0.40	0.43	0.57	0.46	0.36	0.32	0.24	0.17	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.55	-0.69	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page